

in V.32 and V.34 modems, the ones used commonly in dial-up or ISDN lines.

Putting an ADSL circuit in place on existing telephone infrastructure has a load of hardware related problems too. Loading coils, for instance, are essentially low-pass filters that cut off all frequencies above 4 KHz. Unfortunately, all digital modems require frequencies higher than 4 KHz. Loading coils must be removed from the telephone network before any digital service can be availed.

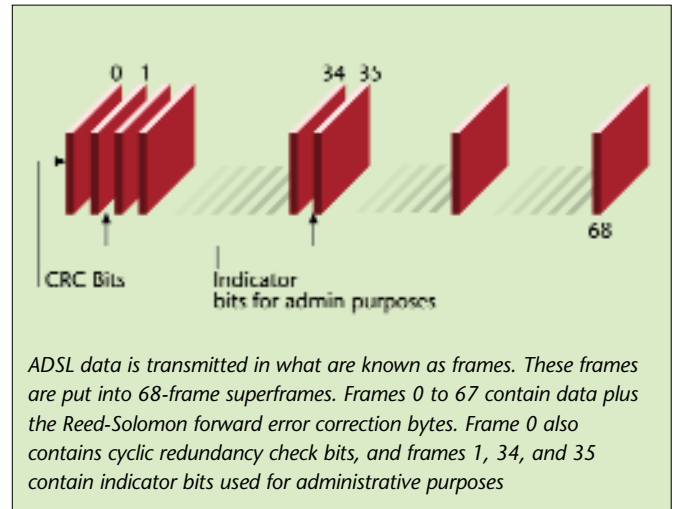
ADSL in India

ADSL attempts in India have been based on the classic copper cable infrastructure with ADSL modems. Dishnet DSL, a Chennai-based company, has been offering Digital Subscriber Lines since the late nineties. A

franchisee-based pilot trial in Kolkata by one operator offering voice, data and video-on-demand was said to be a great success. Services offered include ADSL-Lite and ADSL, which have been introduced in a couple of metros under several tariff packages. These packages range from low monthly subscription, specified per-MB download charge, a charge per view of video-on-demand, to high flat rate monthly rental with unlimited downloads and a low per view charge for video-on-demand.

There are reasons for low connectivity for services such as xDSL in India. Low PC penetration, subscription price, lack of proper infrastructure and security and service-related problems are major contributors to the sluggish growth of xDSL services.

Recently, the Confederation of Indian Industry (CII) finalised a document for the broadband market named Vision 2010. The document envisages India as one of the top five countries in the world in terms of broadband penetration. The Telecom Regulatory Authority of India (TRAI) has also released a paper analysing broadband penetration in India. The CII National Broadband Economy Committee has defined broadband as an always-on network capable of providing interactive voice, data and video services on public networks, reports Cable Quest magazine. Technically, true broadband would be



capable of providing a minimum asymmetric bandwidth of 1.5 Kbps.

In the US, a recent legal ruling abolishing price ceilings and structures for telecom companies in leasing infrastructure services to DSL operators has thrown the establishment into a tizzy. The ADSL market in India, however, recently got a shot in the arm with MTNL announcing its ADSL services in Mumbai and New Delhi. India now has 45,000 DSL broadband subscribers, growing 50 per cent in the latter half of 2003, with only 0.11 per cent penetration of its phone lines. A survey done by Pielle Consulting, London, under the auspices of the DSL Forum, came up with these results. The DSL forum has a mass-market target of 20 per cent penetration of all telephone lines by 2005 in countries where ADSL connectivity already has a base.

Looks like ADSL will at last be a Formula One solution for the 'last mile'.

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Different flavours of DSL

Types of DSL other than ADSL have been deployed, albeit none so widely as ADSL. For example, IDSL is an acronym for ISDN Digital Subscriber Line. The data rate is 128 Kbps across 18,000 feet on 24 gauge wire. It is similar to ISDN service but with data only—there's no voice on the same line. DSL Lite (same as G.Lite) is a 'splitter-less' DSL where the data rate varies from 1.544 Mbps to 6 Mbps downstream, depending on the service subscribed to. The typical range is 18,000 feet on 24 gauge wire. This is the standard ADSL in the sense that it sacrifices speed for not having to install a splitter at the user's home or business. G.Lite is DSL Lite without the 'truck roll', or the cost of setting up the splitter arrangement at the user's end. The splitting is done remotely at the telco's end.



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